



LECTURE 1

WHY GEOSPATIAL REPRESENTATION LEARNING?

Geospatial Representation Learning

Learning Outcomes

Lecture

- Describe the motivation and scope of geospatial representation learning.
- Explain why location, spatial context, scale, and time matter for geospatial data.
- Identify geospatial AI tasks such as classification, prediction, retrieval, mapping, and monitoring.
- Distinguish geospatial representation learning from general machine learning and classical geospatial analysis.
- Recognize geospatial features, embeddings, location encoders, and neural representations.
- Relate the topic to Earth observation, GIS, mobile robotics, agriculture, climate, biodiversity, and urban analysis.

Lab

- Load and inspect a simple geospatial dataset in Python.
- Visualize geospatial observations on a map.
- Describe how spatial location and context influence a machine learning task.
- Extract simple features from geospatial data and interpret them as first representations.
- Document the workflow and observations in a reproducible exercise report.



PRACTICAL 1

EXPLORING GEOSPATIAL REPRESENTATIONS

Geospatial Representation Learning